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6. A class investigated the rate of the reaction between magnesium and hydrochloric acid by measuring the volume of gas produced in 10 seconds using different concentrations of acid.

(a) Tick (✓) the name of the piece of apparatus they used to measure the volume of gas produced. [1]

beaker ☐

thermometer ☐

gas syringe ☐

conical flask ☐

(b) The table shows the results of their experiment.

Concentration of acid (M)	Volume of gas produced (cm^3)			
	Test 1	Test 2	Test 3	Mean
0.2	16	14	15	15
0.4	31	33	30	32
0.6	47	49	29	48
0.8	63	64	65	64
1.0	82	83	79	81

(i) **Circle** the result in the table which was **not** used to calculate a mean.

Give the reason why this result was not used. [2]

Reason ....

....

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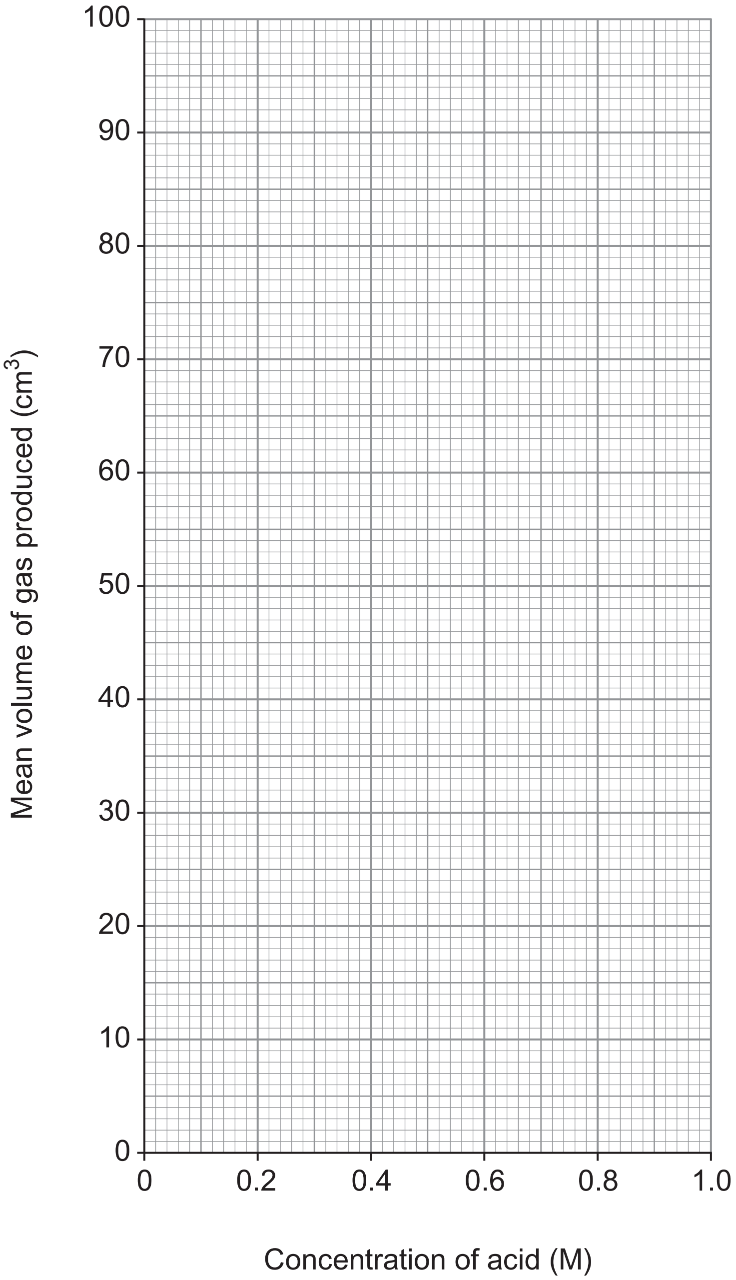
(ii) Plot the mean volume of gas produced against the concentration of acid. Draw a suitable line. [3]



(iii) Use the information to find the volume of gas produced in 10 seconds using acid of concentration 0.5 M. [1]

.... cm^3

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(c) In another investigation the class found that as the temperature of the acid increases the reaction is faster.

Tick (✓) the **two** statements which explain why the reaction is faster at a higher temperature. [2]

there are more particles of acid	☐
the acid particles are moving faster	☐
there is a higher surface area	☐
there are more collisions per second	☐
the acid particles have less energy	☐

(d) The class wanted to identify the gas produced. They tested it as shown in the table.

Test	bubble the gas through limewater	place a burning splint in the gas
Observation	limewater did not change	a squeaky pop and the splint went out

State the name of the gas. [1]

....

10

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